

Darren Zheng

 [Personal Website](#)

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EDUCATION

University of Pennsylvania

Aug 2021 – May 2025

B.A. in Mathematics, Minor in Statistics & Data Science, Ben Franklin Scholar

B.A. GPA: 3.51/4.0

- **Relevant Coursework:** Data Science in Python, Statistical Computing with R, Optimization, Computer Algorithms, Corporate Finance, Probability Theory, Stochastic Processes, Statistical Inference, Partial Differential Equations

PROJECTS

Undergraduate Honors Thesis

June 2024 – May 2025

- Constructed intrinsic Morse functions on Grassmannian manifolds $\mathbf{G}_k(\mathbb{R}^n)$ to compute singular homology groups

Directed Reading Program

Aug 2022 – May 2025

- Engaged in independent study across 3 years, covering knot theory, generating functions, Brownian motion, stochastic calculus, additive combinatorics, Falconer's conjecture, and mapping class groups
- Prepared expository write-ups and delivered presentations each semester to the Mathematics Department, translating abstract concepts to a diverse audience of students and professors

EWS Detection using Topological Data Analysis

Jan 2024 – Apr 2024

- Applied persistent homology techniques to time-series data to develop early warning signals for market crashes
- Implemented Python pipeline (`scikit-tda`, `risper`, `persim`) to transform historic stock index data into persistence landscapes and diagrams for sliding window analysis
- Conducted statistical validation using Mann-Kendall test to analyze significance of methodology

EXPERIENCE

Private Competitive Mathematics Coaching | Tutor

May 2025 – Present

- Guide 14 students with individualized training for math contests, such as AIME, USAMO, and MathCounts
- Prepare personalized problem sets & mock exams to target technical gaps based on past student performance

University of Pennsylvania Math Department | Teaching Assistant

Aug 2022 – May 2025

- Developed and delivered weekly recitations for various proof-based calculus and linear algebra courses
- Graded homework/exams, providing feedback on proof-writing; Received 2 Departmental teaching awards

Ross Mathematics Program | Administrator

June 2024 – July 2024

- Managed operations for a 6-week residential summer math camp for ~ 80 high schoolers and ~ 20 instructional staff
- Scheduled social and educational programming to minimize time conflicts, working alongside the host university
- Presented mathematical talks, involving formal logic, algebraic and point-set topology to advanced students

AwesomeMath Summer Program | Teaching Assistant

June - Aug 2022/2023/2025

- Instructed six 3-week-long courses of ~ 15 -30 students, facilitating problem sessions and active learning
- Taught primarily number theory, combinatorics, and abstract algebra

ACTIVITIES

Penn Math Contest Club | Co-founder & President

Aug 2021 – May 2025

- Established a new student club that organizes an on-campus high-school mathematics contest with original problems
- Planned event logistics, coordinating with campus facilities & training ~ 30 club members/volunteers; Secured corporate sponsorships to offset operational expenses and created handbooks for future leadership
- Grew from ~ 40 online contestants (COVID) to ~ 170 in-person and ~ 200 online in Spring 2025 through outreach

Penn Undergraduate Math Society | President

Jan 2024 – May 2025

- Led Penn's undergraduate math club, directing a board of ~ 10 to support math-adjacent programming
- Organized multiple events (~ 13 per semester), including major/course advising, research talks, integration bee, merch sale, faculty lunches, DRP x PUMS collaboration, and industry panels

Science Olympiad at Penn | Test Writer & Event Supervisor

Aug 2021 - May 2025

- Authored 7 state-level tests over 4 years and designed practical lab components, specializing in chronology, astronomy, cryptography, optics and Fermi estimates
- Utilized automated rubrics and guidelines for volunteers to ensure consistent scoring across 200+ competitors

Student Activities Council | Executive Board

Feb 2023 – Dec 2024

- Oversaw strategic allocation of an ~ 1.7 million annual budget to over 300 student groups, determining budgets and representing their interests to the University administration
- Spearheaded changes to club debt policies and interviews for the new group recognition process

TECHNICAL SKILLS

Languages: Python, Java, SQL, Go, JavaScript

Tools: NumPy, Pandas, R, MATLAB, $\text{\LaTeX}$, Excel VBA, HTML/CSS, Git